

Ryan Cho

Senior Software Engineer — AI Applications & Backend Systems

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PROFESSIONAL SUMMARY

Software engineer with 8+ years building backend services, AI-enabled applications, and platform-scale systems across finance and enterprise environments. Experienced architecting Python and Java microservices, designing LLM workflows with retrieval-augmented generation, and shipping secure APIs that meet regulated compliance requirements. Known for translating complex stakeholder needs into reliable production systems, mentoring engineers, and optimizing performance under rapid growth.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and deployed **Python-based AI agent services** integrating **large language models** to automate document classification and portfolio reporting workflows, reducing manual analyst review cycles across **40+ internal stakeholders**
- Built **retrieval-augmented generation pipelines** using **vector embeddings** and structured fund data to power contextual Q&A tools for investment research teams
- Designed **RESTful microservices** in **Java Spring Boot** and Python serving real-time analytics APIs consumed by web and mobile clients across distributed engineering groups
- Led **SQL query optimization** and schema design in **PostgreSQL**, improving report generation latency for high-volume transactional datasets through indexing and batch retrieval refactors
- Established **regulated-data security controls** including encryption at rest, audit logging, and role-based access for systems handling sensitive client financial information
- Implemented **modular context orchestration layers** enabling tool invocation across internal AI experimentation platforms and streamlining integration with downstream workflow services
- Mentored **3 junior engineers** on microservice design, prompt engineering practices, and production monitoring, elevating team delivery consistency during platform expansion
- Partnered with product and operations leaders to translate **ambiguous business requirements** into technical specifications and iterative prototypes for workflow automation features
- Owned **on-call rotation** and production observability for AI-backed services, diagnosing latency spikes and scaling containerized workloads to support **3x traffic growth**

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Java microservices** on **Spring Boot** supporting trade processing and risk reporting pipelines for institutional banking operations
- Optimized **SQL stored procedures** and **PostgreSQL** queries powering regulatory dashboards, improving batch job reliability for daily compliance submissions
- Built internal APIs integrating legacy mainframe data with modern **REST endpoints**, enabling cross-team consumption by front-office and technology groups
- Collaborated with non-technical stakeholders to document requirements and deliver incremental releases for **workflow automation** tools used by operations analysts
- Implemented unit and integration test coverage for backend services, strengthening deployment confidence across **CI/CD pipelines** in regulated environments
- Supported production incident resolution and performance tuning for high-throughput transaction services during peak market windows

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Contributed to **full-stack features** using **React** and **Node.js** for restaurant inventory management, delivering responsive UI components aligned with product specifications
- Integrated **REST APIs** with PostgreSQL-backed services for real-time stock tracking across multiple client deployments
- Participated in code reviews and sprint planning, gaining hands-on experience with **agile delivery** in a remote engineering team
- Resolved production bugs in authentication and data sync flows, improving reliability for onboarding new restaurant partners
- Assisted in writing automated tests and deployment scripts supporting **continuous integration** workflows

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built responsive **web pages** and donation flows for nonprofit campaigns using **HTML, CSS, and JavaScript**
- Integrated third-party payment APIs to process contributions securely for foundation fundraising initiatives
- Collaborated with designers to implement **mobile-friendly layouts** improving accessibility across donor touchpoints
- Maintained content updates and bug fixes for foundation website during remote contract engagement

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML/CSS, Responsive UI

Backend: Python, Java, Spring Boot, Node.js, REST APIs, Microservices, LLM Integration, RAG, Prompt Engineering

Databases: PostgreSQL, MySQL, SQL, Query Optimization, Vector Databases

Cloud & DevOps: Docker, Kubernetes, AWS, CI/CD, Container Orchestration

Monitoring & Tools: Datadog, Prometheus, Grafana, Git, API Design, Context Orchestration

Workflow & PM: Agile/Scrum, Jira, Cross-functional Collaboration, Technical Mentorship, Stakeholder Communication